

Experiment – 15

CSA1717 - Artificial Intelligence

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Aim:

To write a Python program to implement list operations such as nested list, length, concatenation, membership, iteration, indexing, and slicing.

Algorithm:

1. Start the program.
2. Create a list with elements and a nested list.
3. Display the nested list.
4. Find the length of the list using len().
5. Concatenate two lists using +.
6. Check membership using in.
7. Iterate through the list using a for loop.
8. Access elements using indexing.
9. Display a portion of the list using slicing.
10. Stop the program.

Source Code:

```
# Program to implement list operations
```

```
# 1. Nested List
```

```
nested_list = [1, 2, ["apple", "banana"], [10, 20, 30]]
```

```
print("Nested List:", nested_list)
```

```
# 2. Length of List
```

```
print("Length of nested_list:", len(nested_list))
```

```
# 3. Concatenation
```

```
list1 = [10, 20, 30]
list2 = [40, 50]
concat_list = list1 + list2
print("Concatenation (list1 + list2):", concat_list)
```

4. Membership

```
print("Is 20 in list1?", 20 in list1)
print("Is 100 in list1?", 100 in list1)
```

5. Iteration

```
print("Iterating through list1:")
for item in list1:
    print(item)
```

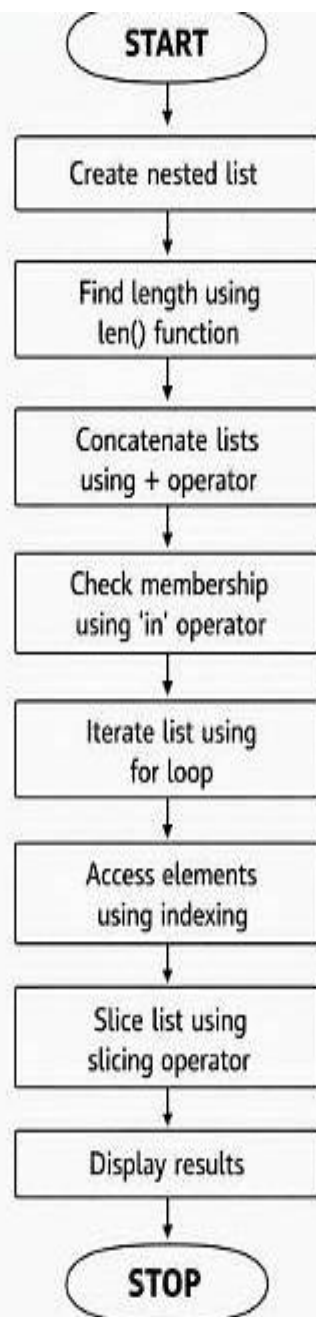
6. Indexing

```
print("Element at index 0 in list1:", list1[0])
print("Element at index 2 in nested_list:", nested_list[2])
```

7. Slicing

```
print("Slicing list1 (index 1 to 3):", list1[1:3])
print("Slicing nested_list (index 1 to end):", nested_list[1:])
```

Flowchart:



Output:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit  
Enter "help" below or click "Help" above for more information.  
>>>  
= RESTART: C:/Users/hemal/OneDrive/Documents/SIMATS/AI/AI_PY/EXP 15_code_AI.py  
Nested List: [1, 2, ['apple', 'banana'], [10, 20, 30]]  
Length of nested_list: 4  
Concatenation (list1 + list2): [10, 20, 30, 40, 50]  
Is 20 in list1? True  
Is 100 in list1? False  
Iterating through list1:  
10  
20  
30  
Element at index 0 in list1: 10  
Element at index 2 in nested_list: ['apple', 'banana']  
Slicing list1 (index 1 to 3): [20, 30]  
Slicing nested_list (index 1 to end): [2, ['apple', 'banana'], [10, 20, 30]]
```

Result:

The program successfully demonstrates list operations such as nested list creation, length calculation, concatenation, membership testing, iteration, indexing, and slicing.